MTH321 · Numerical Analysis of ODEs and PDEs

Numerical Solution of the Robertson Kinetics IVP

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Stiff chemical kinetics · implicit Euler + Newton · step-doubling
No closed form · BDF/Radau oracle · cost = RHS evaluations

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Presentation Roadmap

1. **Motivation**
   Why is this problem hard?
2. **Model & Methods**
   What do we solve, and how?
3. **Stability**
   Can we predict the step-size limit?
4. **Validation**
   Can we trust the numbers?
5. **Efficiency & Insight**
   Who is cheapest at matched error?

What Makes This Problem Hard?

Stiffness here = a stability restriction, not an accuracy restriction

k1	k3	k2
0.04	**10^4**	**3×10^7**

k2 / k1 = 7.5×10^8 · no closed-form solution

1 No closed form

Needs a verified oracle
(BDF + tightened Radau)

2 Exact invariant, y ≥ 0

Checkable separately
from accuracy E(h)

3 Predictable limit

Explicit bound
h < 2/|λ|_max ≈ 5.90×10^−4

Takeaway | This is a stability problem for explicit methods, not a missing-order problem.

The Robertson System

Mass-action ODE · y(0)=(1,0,0)^T · t ∈ [0, 40]

$$y_1' = -0.04\,y_1 + 10^4 y_2 y_3$$
$$y_2' = 0.04\,y_1 - 10^4 y_2 y_3 - 3\times 10^7 y_2^2$$
$$y_3' = 3\times 10^7 y_2^2$$

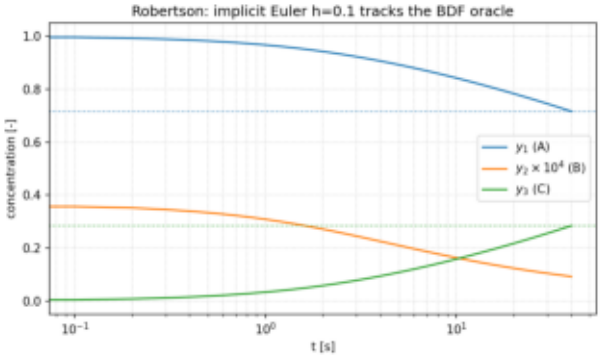
$1^\top f(y) \equiv 0 \Rightarrow y_1 + y_2 + y_3 = 1$
$\Rightarrow$ structural zero eigenvalue of J_f

Stiffness ratio

$$S = \frac{\max_{\lambda\neq 0}|\mathrm{Re}\,\lambda|}{\min_{\lambda\neq 0}|\mathrm{Re}\,\lambda|}$$

At t = 40 (oracle)

$\lambda \approx -3.39\times10^3,\ 0,\ -0.021$
$S(40) = 1.58\times10^5$
Drop the structural zero



Implicit Euler h=0.1 tracks the slow drift (y2 scaled ×10⁴). Not the order proof.

Numerical Methods: Three Integrators + Newton

Test equation $y'=\lambda y$, $z=h\lambda$ · $y_{n+1}=R(z)\,y_n$

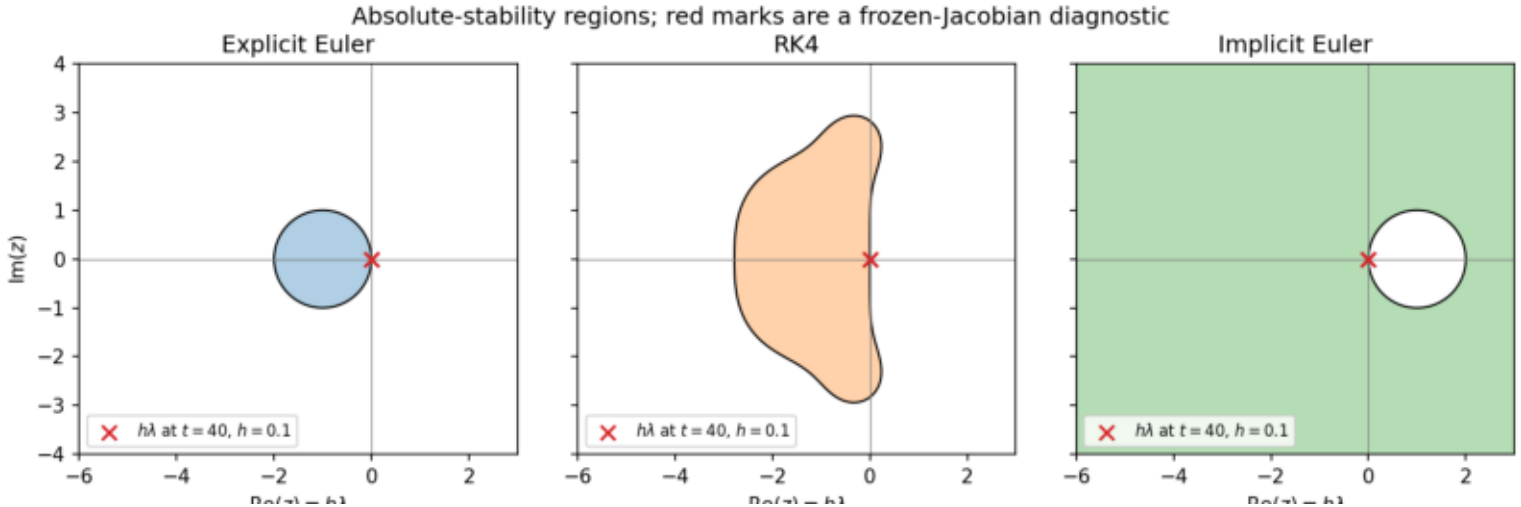
Method	R(z)	Order	Type	Stability
Explicit Euler	$1+z$	1	explicit	bounded disk
RK4	$1+z+z^2/2+z^3/6+z^4/24$	4	explicit	bounded
Implicit Euler	$1/(1-z)$	1	implicit	A- and L-stable

- Implicit stages: Newton on $F(w)=w-y_n-h\,f(t_{n+1},w)=0$, $J_F=I-h\,J_f$, stop $\|F\|_\infty<10^{-12}$.
- Explicit Euler / RK4 need NO Newton (f evaluated at known points only).
- Adaptivity: step-doubling; accept y_{fine} if $e=\|y_{fine}-y_{coarse}\|_\infty \le tol$; else $h\leftarrow h/2$.
- Each accepted adaptive step costs THREE implicit Euler solves (1 coarse + 2 half-steps).
- Oracle: SciPy BDF (rtol=10^{-12}) cross-checked with Radau (rtol=10^{-13}); not a method under test.
- Three errors: $\|\hat{y}-y_{ref}\| \le$ discretisation + Newton + oracle reference.

Takeaway | A-stability, a small Newton residual, and accuracy are three different claims.

Stability Regions

|R(z)|≤1 (shaded) · hλ overlay is a frozen-Jacobian diagnostic, not a nonlinear proof

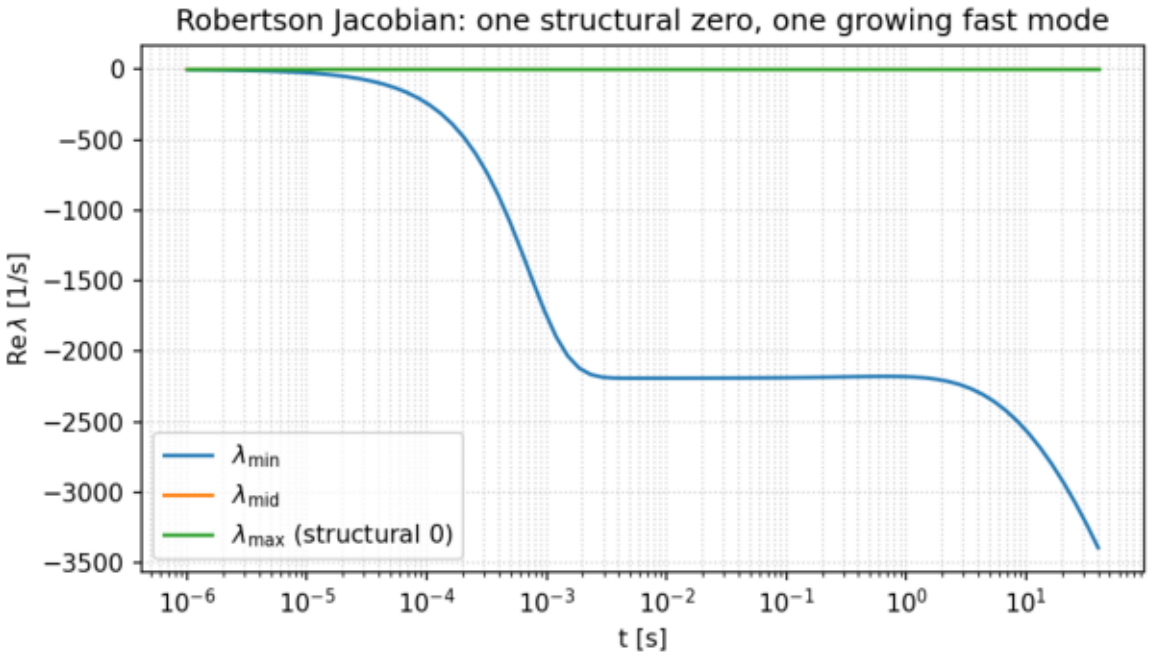


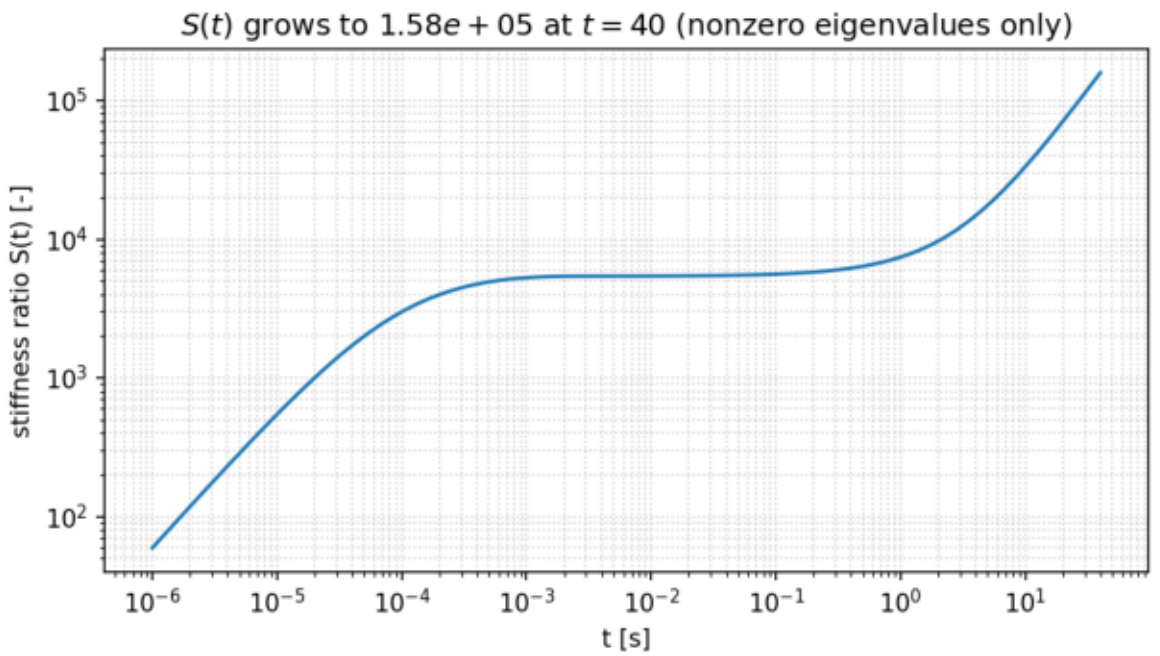
- Explicit Euler: disk |z+1|≤1;
- real axis needs h ≤ 2/|λ|.
- RK4: larger but still bounded.
- Implicit Euler: left half-plane
- (A-stable); R(z)→0 ⇒ L-stable.
- At h=0.1, t=40: fast mode
- z ≈ −339 — outside both
- explicit regions, inside IE.

Takeaway | Bounded ≠ accurate. A-stable means the linear test does not explode; it does not fix the O(h) error.

Eigenvalues and Stiffness Ratio

Along the BDF/Radau oracle · structural zero excluded from S(t)

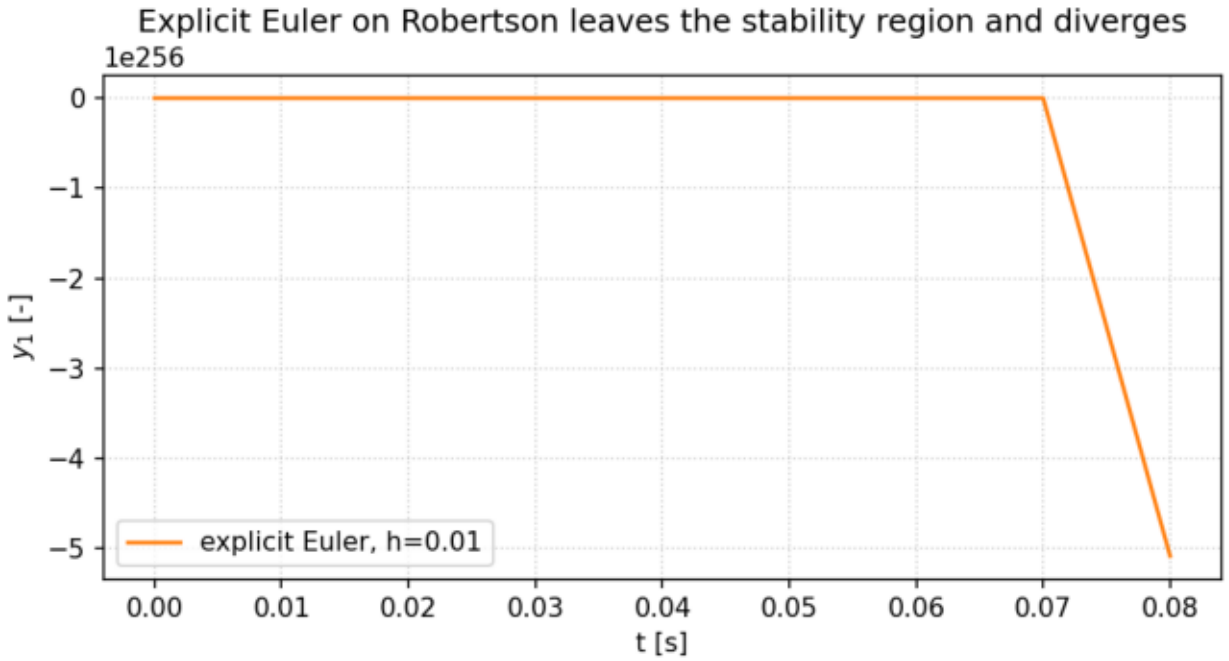




Takeaway | S(40)=1.58×10⁵ from the two nonzero eigenvalues. The fast rate is O(10³), not 3×10⁷.

Testable Step-Size Boundary

Prediction: $h_{\text{crit}} = 2/|\lambda|_{\max} = 2/(3.393 \times 10^3) \approx 5.90 \times 10^{-4}$



Explicit Euler experiment

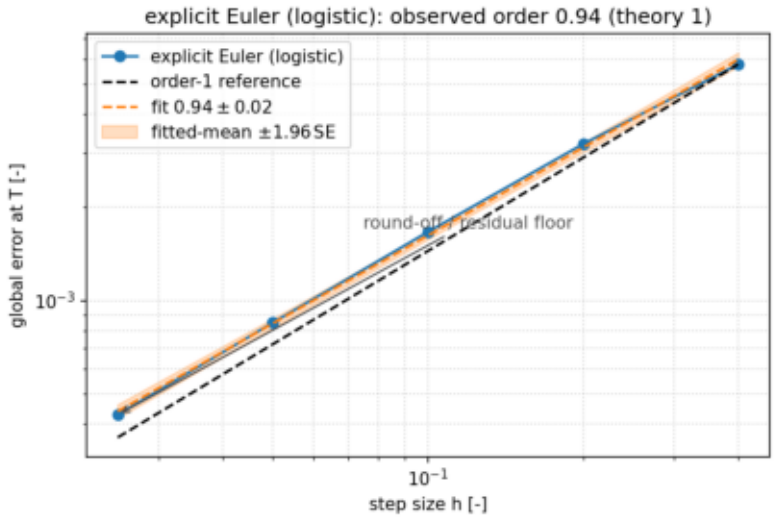
h = 0.01 (≈ 17 × h_crit)
→ Inf / NaN on [0, 40]

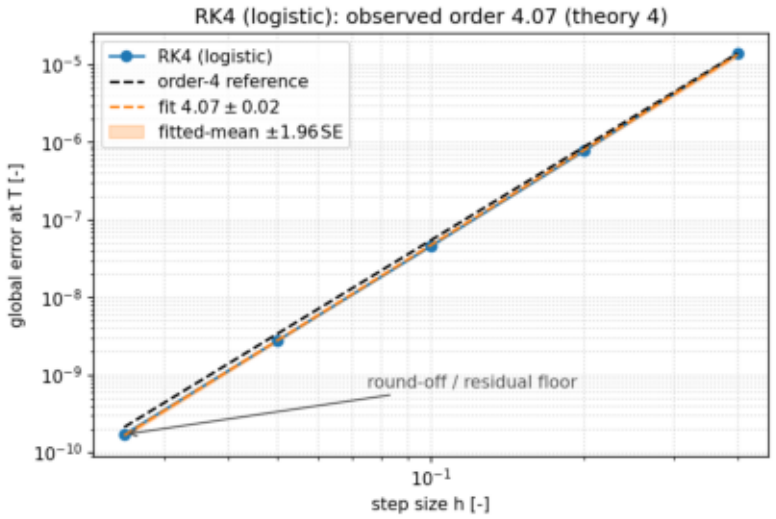
True solution stays on the mass simplex — this is a stability failure, not a large truncation error on a bounded orbit.

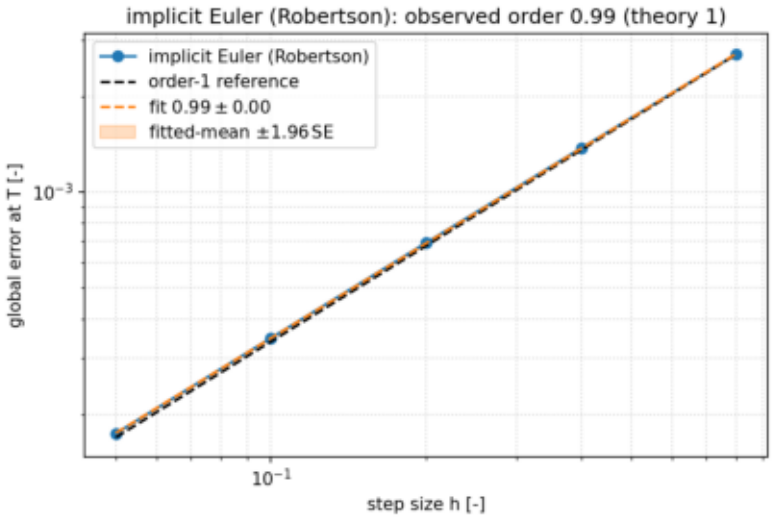
Takeaway | Divergence at h=0.01 matches the frozen-Jacobian bound. Adaptivity does not lift the explicit limit.

Are the Numerical Results Trustworthy?

p ± SE is one standard error, not a 95% CI for p · oracle error ≪ E(h)





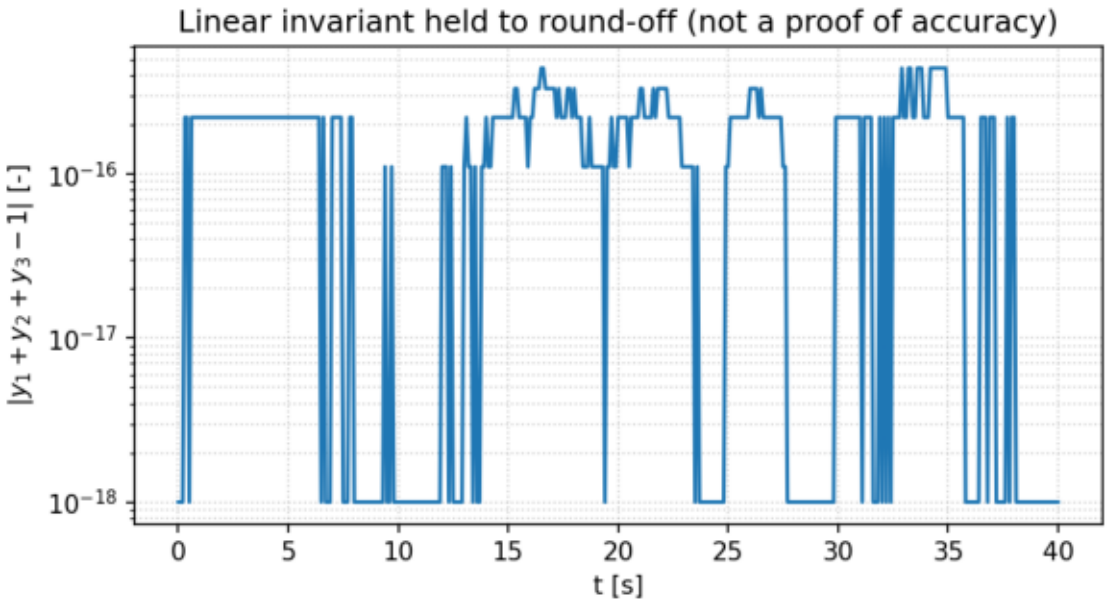


Logistic (exact): Euler 0.943±0.015 (theory 1) · RK4 4.075±0.018 (theory 4)
Robertson vs BDF/Radau: IE 0.990±0.002 · |BDF−Radau|_∞ = 1.4×10^{−11} · y_ref(40)≈(0.715827, 9.186×10^{−6}, 0.284164)

Takeaway | Observed slopes match theory once h is asymptotic. Newton converging ≠ grid converging.

Diagnostics: Three Checks Fail Independently

Conservation · positivity · full-state error E(h) must be reported apart



What the numbers say

Mass defect (h=0.1): 4.4×10^{−16}
→ pass, but a weak test

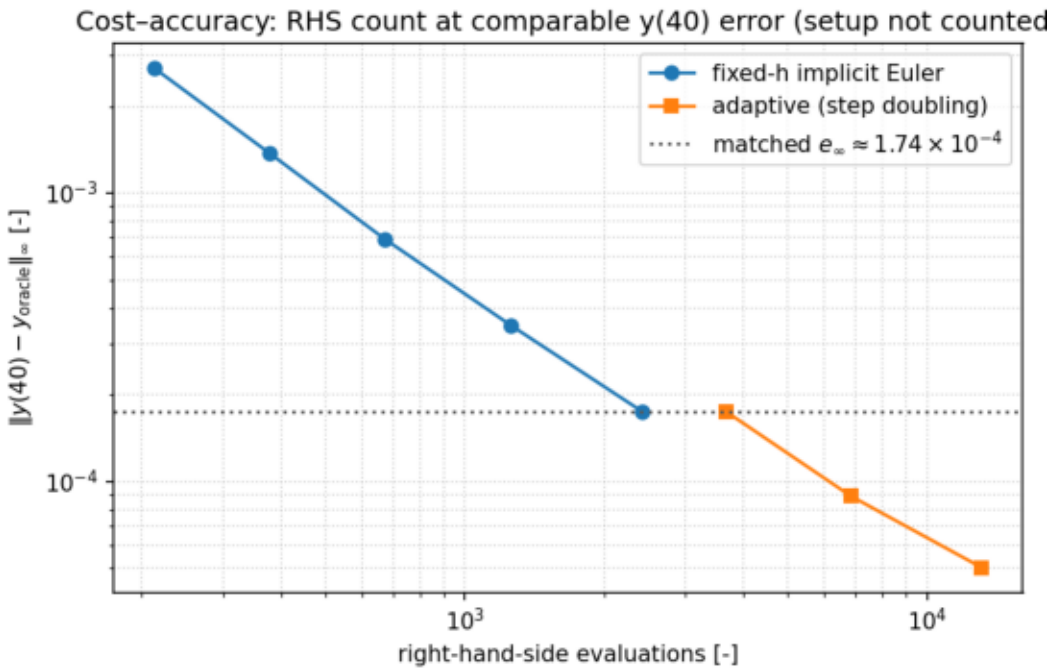
min_{t>0} y2 > 0 for all tabulated h
(even h=10); clip/reject never fired
→ observation, not a theorem

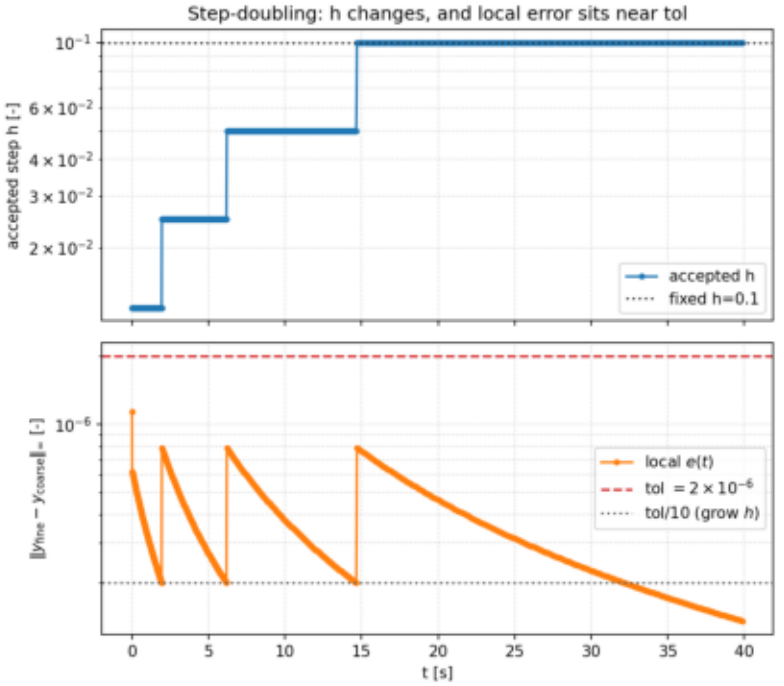
Newton tol 10^{−8} vs 10^{−12}:
Δy(40) = 2.21×10^{−7}
E(0.1) = 3.48×10^{−4}
→ iteration ≪ discretisation

Takeaway | Accuracy is the oracle comparison E(h), not mass, not ||F||, not y2≥0.

Efficiency at a Matched y(40) Error

Cost = RHS evaluations inside Newton · Jacobian factorisation not counted





Takeaway | Fixed h=0.05 → 2424 RHS at e=1.74×10^{−4}; adaptive same error → 3680 RHS. h(t) moves and e sits near tol (749 accept / 3 reject), but step-doubling is more expensive.

What I Learned

1

Results are trustworthy

Orders 0.943 / 4.075 / 0.990; |BDF−Radau|_∞=1.4×10^{−11}; Newton Δy ≪ E(h); mass at round-off.

2

Stiffness = stability restriction

h_crit≈5.90×10^{−4}; explicit h=0.01 diverges. Adaptivity changes h but does not lift the explicit limit.

3

L-stable implicit Euler works

IE is A- and L-stable; at h=0.1 tracks the oracle with componentwise relative errors <1%.

4

Keep diagnostics apart

Conservation, positivity, Newton residual and E(h) fail independently. Adaptive reliability ≠ cheaper RHS cost.

Practical recommendation: use L-stable implicit Euler; drive Newton below y2; monitor E(h), not just mass. Thank you — Questions?